FIELDCOMMAND

Incident Management System v1.0

COMPLETE USER MANUAL

v1.0

ICS/NIMS All-Hazards · Amateur Radio EMCOMM · Public Safety

July 22, 2026

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1 Introduction & System Overview

When a disaster takes down commercial infrastructure — cellular networks, internet, power — the tools modern emergency management depends on disappear precisely when they are needed most. Incident logs revert to paper. Resource tracking becomes a whiteboard. ICS forms are filled out by hand, photocopied, and hand-carried between rooms. Situational awareness degrades to whatever one person can hold in their head. Every organization that has worked a major activation knows this failure mode, and most have simply accepted it as the cost of doing business.

FieldCommand IMS was built to eliminate that failure mode. It is a complete, self-contained incident management platform that carries its own network, its own server, and its own tools — and it operates with no internet connection, no cellular service, and no outside infrastructure of any kind. The cell towers are down, the internet is gone, and you are running on a generator in a parking lot. That is exactly when you need incident management software most — and that is exactly when FieldCommand IMS is designed to perform.

1.1 What FieldCommand IMS Is

FieldCommand IMS is a **complete ICS/NIMS all-hazards incident management system** — not simply an amateur radio tool. It is designed to manage the full lifecycle of any incident from initial response through demobilization, using standard ICS forms and workflows. It runs on a Raspberry Pi 5 server and broadcasts its own private Wi-Fi access point. Any smartphone, tablet, or laptop that connects to that network immediately has access to the full suite of tools through a standard web browser — no app installation, no accounts, and no per-device configuration required.

Version 1.0 provides 48 web-based tools covering every phase of incident management: command setup, incident action plan development, resource tracking, personnel check-in, communications logging, situational awareness, logistics, finance, and demobilization. When internet connectivity is available, live features activate automatically — NWS weather alerts, APRS-IS, animated NEXRAD radar, and HF propagation data. If internet connectivity is lost at any point, all core tools continue without interruption. The system degrades gracefully and recovers automatically.

1.2 How FieldCommand IMS Differs From Other ICS Platforms

Platforms such as WebEOC, E-Team, NIMSIAP, NIMS Logic, and E-iSuite deliver powerful incident management capability — but every one of them assumes a working internet connection and a functioning server infrastructure. They are cloud-dependent by design. When a major disaster disables the very infrastructure those platforms rely on, they go offline with it. The capability disappears at exactly the moment it is most critical.

FieldCommand IMS is built on the opposite assumption: that infrastructure will fail, and that incident management capability must survive that failure. It does this by carrying its own infrastructure — server, network, storage, and tools — in a single deployable package. It does not connect to any external service for core operations. It does not require an IT department to stand up. It does not carry per-seat licensing fees. And it does not fail when the internet fails.

PLATFORM	OFFLINE?	DEPLOYABLE?	NATIVE EMCOMM?	COST
FieldCommand IMS	✓ Fully offline — all features	✓ Deploys in under 30 minutes	✓ Amateur radio + public safety native	Free / open source
WebEOC	✗ Cloud-dependent	✗ Requires server infrastructure	✗ None	$10,000–50,000/year
E-Team	✗ Cloud-dependent	✗ Requires server infrastructure	✗ None	Subscription / agency contract
NIMSIAP	✗ Internet required	✗ Requires connectivity	✗ None	Subscription
E-iSuite	■ Limited offline mode	■ Laptop install — no built-in network	✗ None	License fee

The feature that sets FieldCommand IMS apart from every platform in this category is its native integration of amateur radio and public safety communications directly into the incident management workflow. No other ICS platform includes built-in net control logging, FCC callsign validation against the full national licensee database, APRS tactical mapping, Winlink radio email, JS8Call HF messaging, or AMPRNet gateway capability. These are not add-ons — they are core features, fully integrated with the ICS platform so that radio traffic, net logs, and check-in data flow directly into ICS-309 communications logs, ICS-214 activity logs, and the IAP.

1.3 Key Capabilities at a Glance

CAPABILITY AREA	WHAT IT PROVIDES
ICS/NIMS Incident Management	Full five-section ICS structure — Command, Operations, Planning, Logistics, Finance/Admin. Complete IAP form set including ICS-202 through ICS-221. Live T-card resource board. Planning P cycle tracking. Digital signatures. One-click IAP PDF assembly. FEMA/USCG/NWCG form variants selectable per form.
Amateur Radio EMCOMM	Net control logger with FCC callsign auto-fill from offline database of 800,000+ licensees. Net open/close timestamps, check-out tracking, ICS-309 export. APRS tactical map with live station overlay. Winlink and Pat radio email integration. JS8Call HF digital messaging. AMPRNet (44Net) gateway.
Public Safety Communications	Separate net logger for trunked/P25 radio systems. Radio ID-based check-in. EMA member ID lookup. Interoperable with amateur radio nets on the same incident. ICS-309 export with agency-standard format.
Situational Awareness	Offline APRS tactical mapping. Live NWS weather alerts (when internet up). Animated NEXRAD radar loop with county overlays. GPS-tracked resource map with status-coded markers. HF propagation tool. Barcode/QR check-in scanning.
Resource & Personnel	Member roster with certifications, radio IDs, and barcode check-in codes. QR code generator per member. T-card resource board by type/status/assignment. FEMA PA cost tracking for labor, equipment, and materials. FEMA equipment rate schedule with 44 built-in rates.

CAPABILITY AREA	WHAT IT PROVIDES
Incident Action Plan	Pre-planned event templates for 6 common incident types. FEMA reimbursement documentation with PA cost tracking. Digital signature capture on all forms. One-click IAP PDF compilation. Print center for on-site IAP packages.
WAN & Connectivity	Dual WAN source support — any cellular modem, phone hotspot, or satellite. Preferred/fallback role configuration. Automatic failover. Real-time WAN status dashboard. Configurable detection per source type.
Reference & Offline Content	Kiwix offline Wikipedia and reference library. Radio frequency cheat sheets. Hospital and facilities directory. Repeater database. HF propagation data.

1.4 System Architecture

FieldCommand IMS uses two Raspberry Pi 5 computers. The primary server runs the full application suite and broadcasts the EMCOMM-NET Wi-Fi access point. A dedicated secondary Pi, connected to the same wired switch, serves as an optional AMPRNet (44Net) gateway — maintaining a permanent WireGuard tunnel into the 44.0.0.0/8 amateur radio IP network when internet is available. One or more Wi-Fi routers extend coverage to additional rooms, outdoor staging areas, or upper floors as needed. The entire system can be powered from shore power, battery, or a generator.

COMPONENT	FUNCTION	DEFAULT IP
FieldCommand Pi 5	Primary application server · all 48 web tools · 12+ background services · RAID 1 NVMe storage · EMCOMM-NET AP	192.168.50.1
44Net Gateway Pi 5 (optional)	AMPRNet WireGuard tunnel · callsign-authenticated access · Part 97 access log · isolated from primary server	192.168.50.2
Wi-Fi router (primary)	DHCP server · AiMesh controller · dual WAN management · EMCOMM-NET SSID	192.168.50.254
Wi-Fi mesh nodes (optional)	EMCOMM-NET coverage extension · seamless roaming · same SSID as primary	Assigned by DHCP
Operator workstations	Any device with a modern browser — smartphones, tablets, laptops, Raspberry Pi 500 desktops	Assigned by DHCP

■ The default IP address of the FieldCommand server is 192.168.50.1. This address is configurable during installation. All documentation examples use the default. If your deployment uses a different address, substitute it wherever 192.168.50.1 appears.

2 Getting Started — Connecting to FieldCommand

http://192.168.50.1

Every tool in FieldCommand IMS is a web page served from the Pi at 192.168.50.1 (or your configured address). No app installation is required on any device. Any modern browser works — Chrome, Firefox, Safari, Edge, or the built-in browser on Android or iOS.

Connecting for the First Time

1. Power on the FieldCommand Pi and wait approximately 45 seconds for all services to start. The status LED on the Pi case (if fitted) will turn solid when the server is ready.
2. On any smartphone, tablet, or laptop, open the Wi-Fi settings and connect to the **EMCOMM-NET** network. The default password is printed on the equipment case label and in the Installation Guide. No other credentials are required.
3. Open a browser and navigate to **http://192.168.50.1**. The FieldCommand dashboard loads immediately. If the page does not appear, confirm you are connected to EMCOMM-NET and not to a different network.
4. Bookmark the dashboard URL. On a smartphone, use **Add to Home Screen** to create a shortcut — the web app is designed to work like a native app once bookmarked.
5. From the dashboard, select the tool you need. All tools are accessible from a single page organized by function. No login is required for any tool.

Dashboard Modes

The dashboard presents three selectable modes that filter which tools appear prominently based on your current role. You can switch modes at any time — switching does not affect any data and all tools remain accessible.

MODE	TOOLS SHOWN	BEST FOR
All-Hazards ICS	Full ICS platform, IAP tools, resource tracking, personnel check-in, FEMA cost documentation, event templates	Active incidents requiring ICS structure — any scale or type
Amateur Radio EMCOMM	Net control logger, FCC callsign lookup, APRS map, Winlink, JS8Call, AMPRNet gateway, propagation	Amateur radio net operations, ARES/RACES activations
Public Safety	Starcom/trunked net logger, radio ID roster, resource map, WAN status, weather radar	Public safety radio net operations, Starcom check-in nets

WAN Status Indicators

The top of the dashboard always displays a WAN status bar showing internet connectivity and, if configured, AMPRNet gateway status. The color and label change automatically every 30 seconds.

INDICATOR	MEANING
■ Green — Cellular	Primary internet source (cellular modem, hotspot) is active

INDICATOR	MEANING
■ Blue — Satellite	Satellite internet is active (cellular has failed over)
■ Red — Offline	No internet — all core tools remain fully operational
■ AMPRNet Connected	44Net WireGuard tunnel is up and routing
■ AMPRNet Offline	44Net tunnel is down or gateway Pi is unreachable

■ A red WAN indicator means internet-dependent features (NWS radar, APRS-IS, HF propagation, FCC database updates) are paused. It does not mean FieldCommand IMS is unavailable — all ICS tools, net loggers, roster, forms, and local map features continue operating normally.

3 The Main Dashboard http://192.168.50.1/index.html

The dashboard is the home screen and navigation hub for all FieldCommand tools. It loads automatically when you navigate to the server address. Every tool card on the dashboard launches the corresponding page in the same browser tab — the back button or the navigation bar returns you to the dashboard.

Dashboard Layout

SECTION	CONTENTS
WAN & AMPRNet status bar	Live connectivity status — polls every 30 seconds. Cellular, satellite, or offline indicator. AMPRNet gateway dot.
Mode selector	Three tabs: All-Hazards ICS · Amateur Radio EMCOMM · Public Safety. Filters which tool cards appear in the main grid.
Tool card grid	One card per tool. Color-coded by section: blue = operations, amber = public safety, purple = ICS forms, green = reference. Click any card to launch that tool.
Quick actions	Bottom strip: current incident name, active period, and links to the most recent net log and the print center.

Offline vs. Online Features

The following table shows which features require internet connectivity and which operate fully offline. All core incident management functions are offline.

FEATURE	OFFLINE ?	WHAT CHANGES WHEN OFFLINE
ICS platform (all five sections)	✓ Full	Nothing — fully offline
IAP form set (all ICS forms)	✓ Full	Nothing — fully offline
T-card resource board	✓ Full	Nothing — fully offline
Net control loggers (both)	✓ Full	Nothing — fully offline
FCC callsign lookup	✓ Full	Uses local SQLite database — no internet needed
Member roster	✓ Full	Nothing — fully offline
Offline tactical map (APRS)	✓ Full	Map tiles local — RF APRS still works
Resource map (GPS)	✓ Full	Nothing — fully offline
Kiwix reference library	✓ Full	Nothing — fully offline
NWS weather alerts	■ WAN	Alerts pause — last alert cached and shown
NEXRAD animated radar	■ WAN	Offline overlay shown; radar tiles unavailable
APRS-IS live feed	■ WAN	RF APRS still works; internet feed pauses
HF propagation data	■ WAN	Last retrieved data shown until refresh

FEATURE	OFFLINE ?	WHAT CHANGES WHEN OFFLINE
AMPRNet / 44Net tunnel	■ WAN	Tunnel drops; local EMCOMM-NET unaffected

4 Member Roster http://192.168.50.1/roster.html

The Member Roster is the central personnel database for your organization. It stores every member, mutual-aid visitor, and regular participant — with their callsign, radio ID, certifications, equipment capabilities, and a unique check-in code that enables rapid QR/barcode check-in at activations.

Roster Fields

FIELD	PURPOSE
Member ID	Your organization's internal identifier (e.g. ESV-042)
Callsign	FCC amateur radio callsign — leave blank for non-licensed members
Radio ID	Public safety or trunked radio ID if applicable
Barcode ID	QR/barcode check-in code — defaults to Member ID; can be overridden with a facility badge number or custom code
Name	First and last name
Role	Primary function (Operator, Net Control, Logistics, etc.)
Member type	Member · Visitor · Mutual Aid — affects check-in form behavior
Certifications	ICS-100/200/300/400/700/800, EmComm levels 1–2, CPR, First Aid, CERT
Equipment	HF, VHF, digital modes, APRS, Winlink, go-box, generator, vehicle
Phone / Email	Contact information — stored locally, never transmitted externally
Grid square	Maidenhead grid locator for RF planning

Adding Members

1. Click **+ Add Member** at the top of the roster page.
2. Enter the member's ID, name, and at minimum one identifier — callsign, radio ID, or member ID. All three can be filled if applicable.
3. Select certifications and equipment capabilities using the checkboxes.
4. Click **Save**. The member appears immediately in the roster list.
5. To generate a QR check-in code for the member, click the **QR** button on their roster card. The code can be printed or saved as a PNG for the member to store on their phone.

Importing Members via CSV

If your organization maintains a member list in a spreadsheet, you can import it directly using the ■ **Import CSV** button. The CSV must contain at minimum a **member_id** column. All other columns are optional and will be skipped if not present. Existing members with the same member_id are updated; new member IDs are added.

```
Required CSV column header: member_id
Optional headers: callsign · radio_id · first_name · last_name · role · phone · email · grid
```

■ The roster is stored in the local SQLite database on the Pi. It is included in all automatic backups. Exporting the roster as CSV (■ Export CSV) is recommended before any software update.

5 Operator Identity System http://192.168.50.1/roster.html

The Operator Identity System provides printed and digital identification cards for personnel at activations and exercises. Cards are generated from the roster database and can be printed on standard paper and laminated, or displayed digitally on a phone or tablet.

What an Identity Card Shows

ELEMENT	CONTENT
Name	Full name from roster
Member ID	Organization member number or visitor ID
Callsign / Radio ID	Amateur callsign, public safety radio ID, or both
Role	Primary function at this activation
Certifications	ICS levels and special certifications as colored badges
Organization	Deploying agency or club name
QR check-in code	Scannable code for instant check-in at future activations

Generating Identity Cards

1. Navigate to the roster page and locate the member.
2. Click the member card to open the detail view.
3. Click ■ **Print ID Card**. A print-ready card opens in a new tab.
4. Print on card stock (Avery 5392 or similar) or standard paper for lamination. Cards are sized to fit a standard ID badge holder (3.375" × 2.125").
5. For walk-in mutual-aid personnel without a roster entry, use **+ Walk-In Check-In** on the roster page to create a temporary record and generate a one-time card.

QR Codes for Rapid Check-In

Every member can carry a personal QR code that enables instant check-in at any activation using the Scan Check-In page. The QR button on each roster card opens a modal with the member's personal code. When internet is available, the code is rendered as a true QR image. When offline, a large-text fallback displays the scannable ID for manual entry. Both the QR image and the text code can be printed or saved as a PNG.

6 Amateur Net Control Logger http://192.168.50.1/netcontrol.html

The Amateur Net Control Logger provides a complete net management interface for licensed amateur radio operations including ARES, RACES, AUXCOMM, and any other amateur net. It replaces paper net logs with a live digital record that feeds directly into the ICS-309 communications log for the incident action plan.

Opening a Net

1. Navigate to **Net Control Logger** from the dashboard.
2. Enter the net name, frequency, and mode (SSB/FM/Digital/Other).
3. Select the net type: **Amateur** (for ARES/RACES/AUXCOMM nets).
4. Click **Open Net**. The net opens with a timestamp. The net status banner turns green and shows the elapsed time.
5. The page URL now contains the net ID — share this URL with the ■ **Observer Link** button so section chiefs or served agencies can monitor the net in read-only view on their own devices.

Logging Check-Ins

When a station checks in, type their callsign in the check-in box and press Enter. FieldCommand IMS looks up the callsign instantly in the offline FCC database and fills in the operator's full name and license class automatically. If the station is on your roster, their EMA or member ID is filled as well. No internet is required for callsign lookup — the full national database of 800,000+ amateur licensees is stored locally on the Pi.

1. Type the station's callsign in the **Callsign** field. The name fills automatically from the FCC database as you type.
2. Enter their location or remarks if desired.
3. Click **Check In** or press Enter. The entry appears in the log with a timestamp.
4. To check a station out, click **Check Out** on their row. The checkout time and participation duration are recorded automatically.
5. When a station passes traffic or sends a message, use the traffic entry at the bottom of the page to record it with precedence and addressee.

Closing a Net and Exporting the Log

1. Click **Close Net**. Any stations still checked in are automatically checked out with the net close time. Net duration appears in the header.
2. Click ■ **Export ICS-309** to download the complete ICS-309 Communications Log as a formatted document ready for the incident action plan.
3. The ICS-309 includes: net name, frequency, mode, net open/close times, total duration, and a full check-in table with individual participation durations rounded up to the nearest quarter-hour — the standard for reimbursement documentation.

> ■ The Dead Man's Switch (page: deadmans.html) monitors net activity and sounds an audible alert if no check-in is logged within a configurable time threshold. It is designed for safety monitoring during search and rescue and other field operations where radio silence can indicate an emergency. See Chapter 12 for Dead Man's Switch operation.

7 Public Safety Net Logger http://192.168.50.1/starcom.html

The Public Safety Net Logger handles trunked radio, P25, and other public safety radio systems where check-in is by radio ID rather than amateur callsign. It is designed for interoperability exercises, served-agency support, and any net where participants may not hold amateur licenses.

Key Differences from the Amateur Logger

FEATURE	AMATEUR NET LOGGER	PUBLIC SAFETY NET LOGGER
Primary identifier	FCC callsign — auto-filled from local database	Radio ID (unit number or talk group ID)
License check	FCC ULS lookup — confirms active license	None required — any radio user can check in
Roster lookup	By callsign → returns name and member ID	By radio ID → returns name and member ID
ICS-309 format	Callsign in station column	Radio ID in station column
Typical use	ARES/RACES/AUXCOMM amateur nets	Public safety agency check-in, interop exercises, Starcom-type nets

Operating the Public Safety Net Logger

Operation follows the same workflow as the Amateur Net Logger. Open the net, enter radio IDs as stations check in, and export the ICS-309 at net close. The key operational note is that check-in by radio ID is the standard — operators do not need callsigns, and the FCC database lookup is not invoked. If a roster member has both a radio ID and a callsign, the system will display both on their check-in row for reference.

Observer Mode

Both net loggers support Observer Mode — a read-only, auto-refreshing view of the active net accessible from any device on EMCOMM-NET. The Observer link is generated by clicking ■ **Observer Link** and can be sent by Winlink, JS8Call, or read over the air. Observers cannot modify the log — they see the live net status, check-in list, and traffic log, updating every 15 seconds automatically. No login is required.

> Participants in public safety nets who also hold amateur licenses should check into the public safety net by radio ID only — this maintains the integrity of the agency's radio log. If the same person needs to participate in an amateur net on the same incident, they check into the Amateur Net Logger separately by callsign.

8 Observer Mode — Read-Only Net View http://192.168.50.1/observer.html

Observer Mode provides a read-only, auto-refreshing view of any active net. It is designed for served-agency personnel, EOC staff, emergency managers, and section leaders who need to monitor radio traffic without touching the net control interface. No login, no identity prompt, no setup required.

What Observer Mode Shows

ELEMENT	DESCRIPTION
Net name & status	Active net name, type (Amateur/Starcom), and open/closed indicator
Live check-in log	All stations or units checked in, with timestamp, location, and remarks
Traffic log	Formal message traffic passed during the net
Station count	Live count of checked-in stations at the top of the page
Auto-refresh	Page refreshes automatically every 15 seconds — no manual reload needed

How to Open Observer Mode

1. On the Net Control Logger or Starcom Logger page, click ■ **Observer Link** to copy the URL to your clipboard.
2. Send the link by email, Winlink, JS8Call, or read it over the air to any operator who needs to monitor.
3. The recipient opens the link in any browser on EMCOMM-NET — phones, tablets, and laptops all work.
4. Observer Mode shows the net immediately with no identity prompt and no login.
5. The net name is encoded in the URL so the link can be bookmarked for recurring nets.

> ■ Observer Mode is completely passive — observers cannot add check-ins, log traffic, or close the net. If an observer needs to interact, they must switch to the Net Control Logger or Starcom Logger and identify themselves first.

9 FCC Callsign Lookup http://192.168.50.1/callsign.html

The Callsign Lookup page provides instant offline access to the full FCC Amateur Radio ULS database — approximately 800,000 active licensees. No internet connection is required. Lookups are answered in milliseconds from the local SQLite database on the Pi.

Quick Callsign Lookup

1. Type a callsign (e.g. your callsign) in the search box at the top of the page.
2. Press **Enter** or click ■ **Look Up**.
3. The result shows: full name, license class, expiration date, mailing address, grid square, and FRN.
4. Click **+ Add to Roster** to add the licensee directly to the Member Roster.
5. Recent lookups are saved automatically and shown below the search box for quick re-access.

Advanced Search

1. Click the **Advanced Search** tab.
2. Enter any combination of: First Name, Last Name, City, State, License Class, or Grid Square.
3. Click **Search**. Results show up to 100 matching licensees.
4. Click any result row to see the full record.
5. Click **+ Roster** on any result to add them to the Member Roster.

> ■ The FCC database is refreshed automatically every Sunday at 03:00 via the **fcc-refresh.timer** systemd timer (requires internet connection). The database status — record count and last update date — is shown at the bottom of the page.

10 Tactical APRS Map http://192.168.50.1/tactical.html

The Tactical APRS Map is a full-screen interactive map that combines live APRS station tracking with manual tactical overlays and offline map tiles. It pulls station data simultaneously from Graywolf (port 8080) and YAAC (port 8082), merges the feeds, deduplicates by callsign, and keeps every connected browser updated via WebSocket. The map works completely offline once tiles have been downloaded — no internet connection is needed to see stations or draw overlays.

The default map layer is USGS Imagery+Topo Hybrid — public-domain satellite imagery with roads, contours, and place names overlaid. Additional tile sets (OpenStreetMap, Esri World Imagery) are available in the layer selector. your county map tiles are included with the FieldCommand installation. Additional county and state tile sets are downloaded using the download_tiles.sh script.

Map Layout

ELEMENT	DESCRIPTION
Left sidebar	Station list — all APRS stations received, with callsign, comment, distance, and last heard time
Map area	Interactive Leaflet map with APRS symbols, custom overlays, and drawing tools
Top toolbar	Layer toggles, tile selector, auto-refresh interval, range ring, and drawing tools
Station labels	Callsign labels on the map — toggleable in settings
Status bar	Station count, GPS status, last refresh time

APRS Sources — Graywolf and YAAC

SOURCE	PORT	DESCRIPTION
Graywolf	8080	Primary APRS client — receives RF APRS via connected TNC or soundcard interface
YAAC	8082	Secondary APRS client — backup receive and alternate TNC support

Station Tracking

1. Click any station in the left sidebar to center the map on that station.
2. Click ■ **Track** to lock the map view on a specific station as it moves.
3. Use the **Station age threshold** setting to hide stations older than N minutes.
4. The search box filters the station list by callsign or comment text.

Drawing Tactical Overlays

1. Click a drawing tool in the toolbar: **Marker**, **Polygon**, **Circle**, or **Route**.
2. Click on the map to place the shape. For polygons and routes, click multiple points and double-click to finish.
3. Click a placed marker or shape to add a label or delete it.
4. Overlays are saved to the server automatically and persist across browser sessions.
5. Click **KML Export** to download the current overlay set as a KML file.

GPS Position

If a USB GPS receiver is connected to the Pi, your current position appears on the map as a blue dot and is used to calculate distances in the station list. The GPS status indicator in the status bar shows the fix quality.

Range Rings

Enter a radius in km in the **Range ring** field and click **Draw Ring** to place a circle on the map centered on your GPS position or the station default coordinates. Useful for visualizing coverage areas.

9.4 Map Layers

LAYER	DESCRIPTION	AVAILABILITY
USGS Topo + Imagery (default)	USGS satellite imagery with roads, contours, and place names overlaid. Best for field operations and SAR.	Offline — tiles stored on Pi
OpenStreetMap	Detailed street map with building outlines, trails, and POIs. Good for urban EOC operations.	Offline — tiles stored on Pi
Esri World Imagery	High-resolution satellite imagery without overlays. Best for aerial reconnaissance and damage assessment.	Online only — requires WAN
NOAA NWS Radar (overlay)	Live NOAA radar precipitation overlay on any base layer. Updates every 5 minutes when WAN is available.	Online only — requires WAN

■ Offline map tiles for your county and surrounding counties are pre-loaded during installation. To download tiles for additional counties or states, run: sudo bash /opt/fieldcommand/scripts/download_tiles.sh --region [region-name]. Each county tile set uses approximately 200-500 MB of storage.

11 Starcom Resource Tracking Map http://192.168.50.1/resmap.html

The Resource Tracking Map is a purpose-built situational awareness tool for Starcom public safety net operations. It shows unit positions on an offline map with status color coding, supports manual placement and drag-and-drop repositioning, and synchronizes with the Member Roster for unit information. Access it from the Starcom / Public Safety dashboard mode.

Placing a Unit on the Map

1. Click **+ Add Unit** in the left panel.
2. Enter the **Radio ID**, **Unit Name**, and **Type** (Law Enforcement, Fire, EMS, Search & Rescue, Communications, EOC, Shelter, Other).
3. Enter a **Latitude** and **Longitude** directly, or click ■ **Pick on Map** to click the map and set the position.
4. Set the **Status** (Available, Deployed, Staging, Out of Service).
5. Click **Save**. The unit appears on the map with its status color.

Status Colors

STATUS	COLOR	MEANING
Available	Green	Unit is on scene and available for assignment
Deployed	Blue	Unit is actively assigned and in the field
Staging	Amber	Unit is staging or en route
Out of Service	Red	Unit is unavailable

Moving Units

1. Click and drag a unit marker on the map to a new position.
2. The unit's coordinates update automatically when you release.
3. Alternatively, click the unit marker, click **Edit**, and update the coordinates manually.

Drawing Zones

1. Click **Draw Zone** to enter zone-drawing mode.
2. Click multiple points on the map to define the zone boundary.
3. Double-click to finish. Enter a zone name and color when prompted.
4. Zones persist across browser sessions and appear on the map for all connected devices.
5. Click **Clear Zones** to remove all zones.

12 Resource Board http://192.168.50.1/resources.html

The Resource Board tracks all personnel, vehicles, equipment, and facilities assigned to an activation. Each resource is displayed as a card showing its current status. A stats strip at the top of the page shows live counts by status across all resource types.

The Five Status States

STATUS	COLOR	MEANING
Available	Green	On scene and available for assignment
Assigned	Amber	Actively tasked with a specific assignment
Staging	Blue	En route or at the staging area, not yet deployed
Out of Service	Red	Unavailable — mechanical, medical, or administrative hold
Demobilized	Gray	Released from the incident and departed

Adding a Resource

1. Click **+ Add Resource**.
2. Enter the **Name/Description** (e.g. "IC-7300 HF Radio", "Unit 12 — Medic", "Generator — 5kW Honda").
3. Select the **Type**: Personnel, Vehicle, Radio, Generator, Antenna, Computer/Tablet, Medical, Shelter/Tent, Repeater, or Other.
4. Set the **Status** and optionally fill in Owner/Callsign, Quantity, Location, Assigned To, Contact, and Notes.
5. Click **Save Resource**. The card appears immediately.

Changing a Resource Status

1. Click the colored **status badge** on any resource card.
2. Each click cycles through the five states in order: Available → Assigned → Staging → Out of Service → Demobilized → Available.
3. The stats strip at the top updates instantly.

Filtering the Board

1. Use the **Type** filter dropdown to show only Personnel, Vehicles, Equipment, etc.
2. Use the **Status** filter to show only Available or only Assigned resources.
3. Filters can be combined — e.g. show all Assigned Personnel only.
4. The stats strip always reflects all resources regardless of the active filter.

13 Dead Man's Switch — Net Inactivity Monitor

http://192.168.50.1/deadmans.html

The Dead Man's Switch (DMS) is a net inactivity monitor. When armed for a specific net, it watches the last-activity timestamp on that net. If no check-in, no traffic entry, and no manual reset occurs within the configured threshold period, the DMS fires: an audible alarm sounds on every browser that has the DMS page open, the page background turns red, and a countdown flashes until the situation is resolved. This gives net control operators an automatic safety net against a silent radio, a dropped connection, or an unattended console during a long activation.

The DMS can be armed independently for each active net running in FieldCommand. A major activation running a Starcom General Net, a Weather Net, and a SAR Net simultaneously can have each net on its own DMS with its own threshold. The DMS page runs in the browser tab where it was opened — keep it visible on a dedicated monitor or the net control operator display.

12.1 How to Arm the DMS

1. Open http://192.168.50.1/deadmans.html in a browser on any EMCOMM-NET device.
2. Select the net you want to monitor from the dropdown. All active nets in FieldCommand appear here automatically.
3. Set the threshold — how many minutes of inactivity before the alarm fires. Typical settings: 10 minutes for a busy net, 15 minutes for a weather or traffic net, 5 minutes for a SAR net where frequent check-ins are expected.
4. Click ARM. The countdown timer starts from the last recorded activity on that net.
5. Leave the DMS tab open and visible. The alarm fires in this tab and browser only. For maximum coverage, open the DMS on a dedicated monitor at the net control position.

12.2 When the DMS Fires

When the inactivity threshold is exceeded the DMS fires immediately. The browser tab turns red, a loud alarm tone plays, and the countdown shows how long ago the last activity was recorded. Any of the following actions resets the DMS and stops the alarm:

RESET ACTION	HOW	NOTES
Station check-in	Any operator checks in via the Net Control Logger	Most common reset — running the net normally keeps the DMS satisfied
Traffic entry	Any operator adds a traffic item to the log	Logging a message or status update counts as activity
Manual reset	Click RESET on the DMS page	Use during quiet periods — briefings, breaks, or scheduled silence
Disarm	Click DISARM on the DMS page	Use when the net is formally closed

12.3 Multiple Nets

Each net in FieldCommand can have its own independent DMS instance. Open http://192.168.50.1/deadmans.html in separate browser tabs, one per net, and set different thresholds for each.

For example: the Starcom General Net at 10 minutes, the RACES HF net at 20 minutes during an extended overnight activation, and the SAR Coordination Net at 5 minutes. Each tab monitors and alarms independently.

> The DMS runs in the browser tab — it does not send alerts to other devices. If the operator closes the tab or the device goes to sleep, the DMS stops. For overnight or extended activations, use a dedicated device on AC power with screen-saver and sleep disabled for the DMS monitor.

14 Pre-Flight Deployment Checklist http://192.168.50.1/preflight.html

The Pre-Flight Checklist is a structured GO / CAUTION / NO-GO deployment readiness assessment. Before any activation, run through the checklist to verify that all critical systems are operational.

Checklist Sections

SECTION	ITEMS COVERED
Power Systems	Shore/generator power, battery backup/UPS, solar panels, fuel supply, power distribution
Communications Equipment	HF radio, VHF/UHF radio, go-kit radios, HF antenna, VHF antenna, repeater access, Winlink, JS8Call
Computing & Software	Pi server boot, FCC database, nginx, API server, health monitor, paper backup forms
Networking	ASUS router, EMCOMM-NET Wi-Fi, DHCP, field device connectivity, UniFi switch
Field Supplies	Operator cards, printed forms, batteries, cables, tools, maps, personal PPE
Coordination	Net frequency confirmed, EOC contact established, served agency notified, ICS structure assigned

Deployment Verdicts

VERDICT	MEANING
■ GO	All required items passed — system is ready for deployment
■ CAUTION	Some optional items failed — review before deploying
■ NO-GO	One or more required items failed — do not deploy until resolved

Exporting the Report

1. Click **Export Report** to generate a printable PDF summary of the checklist.
2. The report includes the verdict, timestamp, and notes for all items.
3. File the report with the incident documentation.

> ■ Run the Pre-Flight Checklist before every activation — even for exercises. It takes less than five minutes and catches configuration issues before they become problems during a real emergency.

15 ICS Platform — Overview http://192.168.50.1/ics/

The ICS Platform is the full incident command and documentation system built into FieldCommand. It covers all five standard ICS sections — Command, Operations, Planning, Logistics, and Finance/Admin — plus an interactive Planning P cycle guide. All five sections share a single active incident record, so changes made in Operations are immediately visible in Planning, and the Logistics comms plan feeds directly into printed IAP packages.

The ICS Platform is accessible by selecting the ICS mode from the dashboard mode bar, or directly at http://192.168.50.1/ics/. All data is stored on the Pi and synchronized in real time across every device on EMCOMM-NET. Section chiefs at different tables in the EOC all see the same incident state simultaneously without refreshing their browsers.

Creating a New Incident

1. From the main dashboard (ICS mode), click ■ **New Incident**.
2. Enter the Incident Name (required), Type, Location, Jurisdiction, Incident Commander, Op Period Duration, and an optional Situation Summary.
3. Click **Create Incident**. The incident becomes the active incident for all ICS platform pages.

ICS Navigation Tabs

TAB	SECTION	PRIMARY USE
■ Command	Command Section	Incident objectives, safety, weather, situation
■ Operations	Operations Section	T-card board, resource assignments
■ Planning	Planning Section	IAP documents, resource status table, period tracking
■ Logistics	Logistics Section	Comms plan, supply tracking, meals
■ Finance	Finance / Admin Section	Cost accounting, time tracking, procurement
■ Planning P	Planning Cycle Guide	Step-by-step ICS planning cycle with forms and attendees

Supported Incident Types

TYPE	TYPICAL USE
Natural Disaster	Severe weather, flooding, earthquake, tornado response
HazMat	Chemical spill, gas leak, radiological incident
Search & Rescue	Lost person, overdue hiker, water rescue
Mass Casualty Incident	Multi-patient medical emergency, transportation accident
Weather Net	Starcom/SKYWARN weather spotter activation
Public Health	Disease outbreak, mass vaccination, shelter-in-place
Other	Any incident not covered by the above types

14.3 Multi-User Real-Time Collaboration

Every device on EMCOMM-NET connected to the ICS Platform sees the same incident state simultaneously. Changes made by the Operations Section Chief on one tablet appear immediately on the Planning Section Chief tablet and the Logistics Section Chief laptop without anyone pressing a refresh button. This real-time synchronization uses WebSocket connections maintained by the ics-platform background service on the Pi.

The recommended workflow for a fully-staffed EOC activation is:

POSITION	DEVICE	ICS SECTION
Incident Commander	Tablet on EMCOMM-NET	ics/command.html — objectives, safety, public info
Operations Section Chief	Laptop on EMCOMM-NET	ics/operations.html — T-card board, resource assignments
Planning Section Chief	Laptop on EMCOMM-NET	ics/planning.html — IAP tracker, situation status
Logistics Section Chief	Tablet on EMCOMM-NET	ics/logistics.html — comms plan, supply requests
Finance/Admin	Laptop on EMCOMM-NET	ics/finance.html — cost tracking, time log
Net Control Operator	Pi 500 workstation	netcontrol.html + starcom.html — radio net logging

16 ICS Command Section http://192.168.50.1/ics/command.html

The Command Section captures the strategic-level information for the incident: incident objectives, safety hazards, weather summary, current situation, accomplishments, and planned actions. This content forms the core of the Incident Action Plan (IAP).

Incident Objectives (ICS-202)

1. To add a pre-defined objective: click the **Select a common objective** dropdown. Choose from 100 common ICS objectives organized in 13 groups.
2. The selected objective populates the text field below. Edit it if needed — fill in any bracketed placeholders such as [frequency], [location], or [time].
3. If no edits are needed, click **+ Add** or press Enter.
4. To add a completely custom objective: type it directly in the text field and click **+ Add**.
5. Objectives are listed above the input field. Click the ✕ next to any objective to delete it.

> The 100-item objectives dropdown is organized into 13 groups: Life Safety, Communications, Resource Management, Incident Command, Search & Rescue, Shelter & Mass Care, Weather & Natural Disaster, HazMat, Mass Casualty, Public Information, Finance & Administration, Demobilization, and Agency-Specific.

Situation Report Fields

FIELD	WHAT TO ENTER
Safety Message	Safety message for this operational period (ICS-208)
Current Situation	Brief description of current incident status and key facts
Accomplishments	What has been achieved so far this operational period
Planned Actions	What operations are planned for the next operational period
Weather Summary	Current and forecast weather affecting operations

17 ICS Operations Section http://192.168.50.1/ics/operations.html

The Operations Section manages the tactical resources assigned to the incident. It uses a T-card style status board where resource cards can be dragged between status columns, mirroring a traditional ICS T-card board.

The T-Card Board

COLUMN	MEANING
Available	Resource is checked in and available for assignment
Assigned	Resource has been given a tactical assignment
Out of Service	Resource is unavailable — mechanical, medical, or other
Staging	Resource is at the staging area awaiting assignment
Released	Resource has been demobilized from the incident

Adding a Resource to the Board

1. Click **+ Add Resource** or **+ Assignment**.
2. Enter the resource name, type, callsign/Radio ID, and initial assignment.
3. Click **Save**. The resource appears in the Available column.

Moving Resources Between Columns

1. Drag a resource card from one column to another to update its status.
2. Alternatively, click the resource card to open its detail view, then change the status in the dropdown.
3. All status changes are saved to the server immediately.

Assignment Details

1. Click any resource card to open the detail panel.
2. Enter or update the **Assignment** (Division/Group), **Supervisor**, **Work Location**, and notes.
3. This information feeds the ICS-204 Assignment List.
4. Click ■ **Save** to commit the changes.

18 ICS Planning Section http://192.168.50.1/ics/planning.html

The Planning Section manages the Incident Action Plan documents, resource status summary table, and operational period tracking. It is where the PSC assembles all the pieces of the IAP before each operational period briefing.

IAP Tracker Tab

1. Click the **IAP Tracker** tab.
2. Each required form (ICS-202 through ICS-208) is listed with a completion checkbox.
3. Check off each form as it is completed and added to the IAP package.
4. The tracker shows the percentage of IAP completion at the top.

Resource Status Summary

1. Click the **Resource Status** tab.
2. The table shows all resources by type with current status counts.
3. Click **+ Add Row** to manually add a resource category.
4. Counts update from the Operations Section T-card board automatically.

Incident Documents

1. Click the **Documents** tab to manage IAP-associated documents.
2. Click **+ Add Document** to link an uploaded reference document to this incident.
3. Enter the document title, form number, and notes.
4. Documents are linked to the incident for archival purposes.

> The ICS-309 Communications Log (http://192.168.50.1/ics309.html) can be linked to any active incident from its own page. See Chapter 22 for details on the ICS-309 form.

19 ICS Logistics Section http://192.168.50.1/ics/logistics.html

Logistics provides all facilities, services, and materials that support the incident. It owns the communications plan and tracks supplies, food, medical, facilities, and check-in.

TAB	HOW TO USE IT
Comms Plan (ICS-205)	Build the radio communications plan. Each row is a channel: function, channel name, frequency, CTCSS tone, mode, remarks. Click **Add Comms Row** to add a channel. Pre-fill with your agency's radio channels.
Supply	Log supply requests: item, category, quantity needed, quantity on hand, priority. A progress bar shows the fill rate. Use **Add Supply**.
Facilities	ICP location, base/camp details, staging area manager, and medical aid station info.
Food / Medical (ICS-206)	Meal service log (meal type, count served, menu) plus medical unit leader, ambulance status, hospital contacts.
Check-In (ICS-211)	Personnel check-in list. Records name, ICS position, agency, and arrival time. Links to the Member Roster for pre-populated entries.

20 ICS Finance / Admin Section http://192.168.50.1/ics/finance.html

Finance/Admin manages all financial aspects of the incident: cost accounting, time tracking, procurement, and administrative records for reimbursement.

TAB	HOW TO USE IT
Cost	Log expenditures: category (personnel, equipment, supply, food, transport), amount, description, vendor, and approver. A running total is shown. Use **Add Cost**.
Time	Log person-hours per operator: name, position, agency, hours worked, hourly rate (0 = volunteer). Used for reimbursement claims. Use **Add Time**.
Procurement	Track purchase orders: item, vendor, PO number, amount, status (Requested, Ordered, Delivered, Closed). Use **Add Procurement**.
Admin	Finance/Admin Section Chief info, billing agency, cost-share type, claim deadline, and supporting documentation notes.

All finance data can be exported to CSV for submission to the agency finance officer or FEMA reimbursement process. Click **Export CSV** on any Finance tab.

21 NTS Radiogram Generator http://192.168.50.1/nts.html

Generate properly formatted ARRL National Traffic System radiograms. The form produces the standard preamble, address block, and text section, and keeps a traffic log.

Creating a Radiogram

1. Click **Auto-fill Number** for the next sequential message number, or enter one.
2. Set the **Precedence**: EMERGENCY, PRIORITY, WELFARE, or ROUTINE.
3. Choose **Handling Instructions** (HX codes) if needed — HXA through HXF.
4. Enter **Station of Origin**, **Place of Origin**, and click **Auto-fill Date/Time**.
5. Enter the addressee name, address, and phone.
6. Type the message **Text** in ARRL all-caps style. The Check (word count) calculates automatically.
7. Enter the **Signature**, then click **Generate Radiogram**.
8. Click **Save to Log** to keep a copy, then **Print** for the paper record.

Handling Instruction (HX) Codes

CODE	MEANING
HXA	Collect landline delivery authorized up to $ amount
HXB	Cancel message if not delivered within X hours
HXC	Report date and time of delivery to originating station
HXD	Report to originating station the identity of station from which received
HXE	Delivering station get reply from addressee
HXF	Hold delivery until (date)
HXG	Delivery by mail or landline toll call not required

22 ICS-213 General Message http://192.168.50.1/ics213.html

The ICS-213 is the standard written message form for inter- and intra-agency communications on an incident. Use it when you need a written record of a request, situation update, or resource order.

Completing an ICS-213

1. Fill in the header: incident name, To, From, position/agency, date/time, and subject.
2. Click **Auto-fill** to populate the current date and time.
3. Type the message in the body.
4. Click **Generate Form** to render a print-formatted version.
5. Use the **Print** button to produce a field-ready paper copy.
6. Click **Save to Log** to keep a copy in the form log.

■ Use the ICS-213 for formal written traffic. For routine spoken radio traffic, use the Net Control log instead.

23 ICS-214 Activity Log & ICS-309 http://192.168.50.1/ics214.html

The ICS-214 is a unit-level activity log required for every ICS section and unit. It records personnel assigned and a timestamped log of activities through the operational period.

Completing an ICS-214

1. Enter the incident name, operational period, unit name, and unit leader.
2. Click **Add Personnel** to list each person assigned to the unit (name, position, agency).
3. For each activity, click **Add Entry**, then **Auto Timestamp** for the current time, and type what happened.
4. Entries accumulate in time order throughout the period.
5. Click **Generate Form** and **Print** for the incident record. **Save Local** keeps a copy on the device.

ICS-309 Communications Log

The companion ICS-309 page (Dashboard → ICS mode → ICS-309 Comms Log) records a formal communications log for the operational period. Each entry has a timestamp, from, to, and message subject. Click **Save to Incident** to file it with the active incident for after-action review.

1. Click **Add (timestamp now)** to add a row with the current UTC time.
2. Fill in From, To, and the message Subject for each entry.
3. Click **Generate Form** to render the printable ICS-309.
4. Click **Save to Incident** to file it in the incident record.

24 Winlink Form Import & Incident Archiving

http://192.168.50.1/winlink-import.html

Winlink Express has its own built-in ICS forms. This chapter explains how to bring that form data into the FieldCommand server so the whole incident is documented and archived in one place — and how to re-print a received ICS-213, ICS-214, or ICS-309 on the Pi.

Step-by-Step Import

1. In Winlink Express, open the ICS form message (sent or received).
2. Right-click the **RMS_Express_Form_*.xml** attachment and save it to your computer.
3. On the Winlink Form Import page, drag the saved XML file onto the drop zone, or click Browse.
4. Click **Parse Form Data**. The page detects the form type and extracts all fields.
5. Review the extracted fields in the editable grid. Fix any fields that need correction.
6. Choose the Incident and Direction (Received/Sent), then click ■ **Archive to incident**.

FORM	RE-PRINT ON PI?	NOTES
ICS-213 General Message	Yes	Re-renders as the Pi's ICS-213 for printing
ICS-214 Activity Log	Yes	Activity and personnel lines split into rows automatically
ICS-309 Comms Log	Yes	Log rows parsed into the comms log
Other Winlink forms	Archive only	Data captured and archived — no ICS re-print

Recommended Workflow

1. Run Winlink traffic in Express as your operators normally do.
2. At the end of each operational period, collect any RMS_Express_Form_*.xml files from operators.
3. Import them via the Winlink Form Import page to complete the incident record.
4. Cross-check the ICS-309 log with the Net Control log for completeness.

25 HF Propagation http://192.168.50.1/propagation.html

The propagation page fetches live solar and geomagnetic data from hamqsl.com (when internet is available) and uses it to estimate current HF band conditions.

Reading the Indicators

INDICATOR	WHAT IT MEANS FOR HF
SFI (Solar Flux)	Higher = better high-band propagation. >130 excellent; 90–130 good; <90 40m/80m only.
Sunspot Number	More sunspots = more solar activity = better HF. Follows the 11-year cycle.
A-Index	Daily geomagnetic activity. <10 quiet (good HF); 10–30 unsettled; >30 disturbed.
K-Index (0–9)	3-hour geomagnetic snapshot. ≤2 quiet; 3–4 unsettled; ≥5 storm (HF degraded); ≥7 severe.
X-Ray Flux Class	Solar flare class. A/B/C minor; M moderate (SID possible); X strong HF blackout risk.
Band Condition Bars	Green = good / Amber = fair / Red = poor for each band from 10m through 80m.

> When the Pi has no internet connection, the propagation page shows the last fetched values with a timestamp. For field activations without WAN, check propagation before departing on a device with internet, then switch to the offline Pi for the activation.

26 Repeater Database http://192.168.50.1/repeaters.html

The Repeater Database displays repeater data with rich filtering. It works in three modes: an offline file you load yourself (most reliable), the server API, or built-in sample data. Real repeater data comes from RepeaterBook.com.

The Three Sources

SOURCE	HOW TO USE IT
Offline File (recommended)	Load a RepeaterBook CSV or JSON export. Drag-and-drop the file onto the drop zone, or click Browse. Data persists in your browser after the first load. This needs no API token and is the most dependable method.
Server API	Pulls from the FieldCommand server if the net manager has run fetch_repeaters.py to download RepeaterBook data. Requires a RepeaterBook API token.
Sample Data	Three placeholder entries (SAMPLE-1/2/3) for testing the interface. Shows a "not a real repeater" warning. Do not use for operations.

Filtering and Sorting

FILTER	OPTIONS
Band	2m, 70cm, 23cm, 6m, 10m, 1.25m
State	IL, WI, IN, IA (default), or any US state
Affiliation	ARES, SKYWARN, RACES, SATERN, EmComm
Mode	FM, DMR, C4FM/Fusion, D-STAR, P25, NXDN
Sort	Callsign, Frequency, City, Distance (nearest first)

■ A systemd timer (repeater-refresh.timer) runs the fetch automatically on the first of each month at 04:00, as long as the token is saved in the repeaterbook.env file and the Pi has internet at that time.

27 Facilities Directory http://192.168.50.1/facilities.html

Maintain a directory of EOC locations, hospitals, shelters, staging areas, and command posts. Each entry stores address, coordinates, radio frequencies, CTCSS tone, contact person, on-site callsign, generator status, ADA access, capacity, and operational notes.

Default Facilities

On first startup, FieldCommand seeds four your county facilities: the your county EOC (Woodstock), Centegra Hospital Woodstock, the your county Fairgrounds staging area, and Centegra Hospital your area. Edit these with your actual operational details.

Managing Facilities

1. Click **+ Add** to create a facility, or click any facility card to view its full detail.
2. In the detail view, click **Edit** to change fields, or **Delete** to remove it.
3. Use **Copy Address** to copy the street address to clipboard for navigation apps.
4. Click **Export CSV** to download the full directory for offline reference.

28 Grid Square Calculator http://192.168.50.1/grid.html

Convert between decimal latitude/longitude and Maidenhead grid squares in both directions, and calculate distance and bearing between two grid squares. Supports 4-character (e.g. EN52) and 6-character (e.g. EN52ab) precision.

Using the Calculator

1. To convert coordinates to a grid square: enter latitude and longitude, click **Calculate from Lat/Lon**.
2. To convert a grid square to coordinates: enter the grid square, click **Calculate from Grid**.
3. To use your current location: click **Use My Location** (requires location permission, or uses the configured/GPS coordinates).
4. To find distance and bearing: enter two grid squares and click **Calculate Distance**.

The map on the page highlights the grid square on a North America outline for quick visual reference. For example, northern Illinois is in grid square **EN52**.

29 Radio Cheat Sheets http://192.168.50.1/cheatsheets.html

A quick-reference set of radio and ICS cheat sheets, organized into tabs. No internet needed — everything is built in.

TAB	CONTENTS
Phonetic Alphabet	NATO phonetics A–Z with pronunciation, plus ITU number pronunciation (Zero, WUN, TOO, TREE, FOW-er, FIFE, SIX, SEV-en, AIT, NIN-er)
Q-Codes	Common amateur Q-codes (QRM, QRN, QRP, QSL, QTH...) and EmComm net Q-codes (QNI, QNS, QND, QNN...)
Prowords	Full procedure word reference — ROGER, WILCO, SAY AGAIN, BREAK, OVER, OUT, CORRECTION, SILENCE, AUTHENTICATE, and more, with meanings and examples
Band Plan	2m and 70cm segments, HF emergency frequencies (ARES, ARRL, IARU, 60m channels), and service bands (MURS, FRS, GMRS, Marine, Aviation)
NTS Precedence	EMERGENCY / PRIORITY / WELFARE / ROUTINE — definitions, when to use each, and examples
ICS Positions	Standard ICS command and general staff titles with abbreviations (IC, OSC, PSC, LSC, FSC, SO, IO, LO)
CTCSS/DCS Tones	Standard CTCSS tone table (67.0–254.1 Hz) and DCS code table for repeater access
Signal Reports	RST and RS signal report codes, S-meter calibration, and signal quality descriptions

30 Print Center http://192.168.50.1/printcenter.html

The Print Center is a one-stop hub for every printable document, plus a built-in incident cover-sheet generator.

CATEGORY	DOCUMENTS
ICS / NTS Forms	ICS-213 General Message, ICS-214 Activity Log, NTS Radiogram, Pre-Flight Checklist
Reference Cards	Phonetic Alphabet, Q-Codes & Prowords, ICS Structure & Forms, CTCSS/DCS & Signal Reports
Operations	Net Control Log (ICS-309), Starcom Net Log (ICS-309), Member Roster, Resource Board
Cover Sheet	Generate a formatted IAP cover sheet from incident details

Generating an Incident Cover Sheet

1. Fill in incident name, number, IC, operational period, frequency, location, and situation summary.
2. Click **Generate Cover** to preview it in the page.
3. Click **Print Cover** to print it in a new window.
4. Click **Clear** to reset the form.

Connecting a Printer to EMCOMM-NET

FieldCommand has print buttons on 17 pages. All printing uses the browser standard print function — it sends the job to whatever printer the operator's browser can reach. Three options are supported:

OPTION	HOW IT WORKS	BEST FOR
A — Own printer	Each operator prints to their own locally connected printer. No Pi setup needed.	Single operator with their own printer
B — USB printer via Pi (CUPS)	USB printer plugged into the Pi. CUPS shares it across EMCOMM-NET. Installed automatically. Admin at http://192.168.50.1:631. Supports Windows, Mac, iOS AirPrint, Android, Chromebook.	Field activations — one shared printer for all operators
C — Network printer	Wi-Fi or Ethernet printer connected directly to EMCOMM-NET. Devices discover it via Bonjour/mDNS automatically.	Sites with an existing Wi-Fi capable printer

Setting Up Option B — USB Printer via CUPS

1. Plug the USB printer into the Pi or powered USB hub.
2. Open **http://192.168.50.1:631** from any device on EMCOMM-NET.
3. Click **Administration** → **Add Printer** and log in with the Pi username/password.
4. Select the printer from Local Printers. Check **Share This Printer**. Click Continue.
5. Select the driver, click Add Printer, then print a test page.
6. Windows: Settings → Printers → Add a printer — it appears automatically on EMCOMM-NET.
7. iOS/iPad: tap Share → Print in any FieldCommand page — AirPrint finds it automatically.
8. Android: install CUPS Print app → add printer at 192.168.50.1 port 631.

Recommended field printers: **Brother HL-L2350DW** (laser, excellent Linux support), **Canon PIXMA TR150** or **HP OfficeJet 200** (battery-powered portable options). Color multifunction laser options for full IAP package printing: **Brother MFC-L3770CDW**, **HP Color LaserJet Pro MFP M479fdw**, or **Canon imageCLASS MF743Cdw** — all support USB, Wi-Fi, and Ethernet. See the Installation Guide Step 8 for full setup instructions and the complete comparison of all three printer connection options.

29.3 Adding the Printer on Each Device

DEVICE TYPE	HOW TO ADD SHARED PRINTER
Windows laptop	Settings → Bluetooth & devices → Printers & scanners → Add a printer. Windows discovers the CUPS printer automatically on EMCOMM-NET via Bonjour.
Mac / macOS	System Settings → Printers & Scanners → click + (Add Printer). Click the Default tab — shared printer appears via Bonjour. Select and click Add.
iPad / iPhone (AirPrint)	No setup required. CUPS + Avahi advertises as AirPrint-compatible. Tap Share → Print in any FieldCommand page and select the printer.
Android	Install the free CUPS Print app from Google Play. Add printer at IP 192.168.50.1, Port 631, Protocol IPP.
Raspberry Pi 500 workstation	Chromium discovers CUPS printers via Bonjour automatically. No additional setup needed — printer appears in the print dialog.

31 Reference Library http://192.168.50.1/refs.html

The Reference Library is a document-management system for all field reference materials — radio manuals, interoperability plans, ICS form packages, SOGs, agency plans, and training documents. Files are stored on the Pi and accessible from any device on EMCOMM-NET.

TAB	CONTENTS
■ Amateur Radio	Radio manuals, frequency plans, ARRL publications, band plans, antenna refs
■ ICS / Emergency Mgmt	SWIC plans, NIMS docs, ICS form packages, agency SOGs, mutual aid plans
■ All Documents	Every uploaded document regardless of category

Uploading a Document

1. Click ■ **Upload Document**. A panel slides in from the right.
2. Drag-and-drop a file or click Browse. Accepts PDF, Word, Excel, PowerPoint, images, KML, ZIP, GPX — up to 200 MB.
3. Fill in Title, Category, Source, Description, Tags, Revision, and Expiry (if relevant).
4. Under **Show on tab(s)**, check Amateur Radio, ICS/EmMgmt, or both.
5. Click ■ **Upload Document**. PDFs get an automatic thumbnail.

Finding Documents

1. Use the Category filter, tag cloud, or sort options (Newest, Title, Most Downloaded).
2. Switch between grid and list view using the view toggle.
3. Click any document to open its detail, then Download, Edit, or Delete.

32 Kiwix Offline Library http://192.168.50.1:8081/

Kiwix serves a complete offline reference library from the Pi on port 8081. Every device on EMCOMM-NET can browse it with no internet connection at all. The content is stored as ZIM files — compressed, self-contained web archives that function like offline snapshots of entire websites. WikiMed gives your medical support personnel the same reference content a physician would look up during a mass casualty event. Wikipedia Mini provides general field reference for briefings. iFixit gives your communications team step-by-step repair guides for radio equipment, laptops, and generators — all without internet.

Kiwix has its own built-in full-text search bar that searches across all installed content at once. Type any medical symptom, procedure, equipment model number, or topic and results appear instantly, fully offline. The library is accessible from the dashboard Kiwix Library card, or directly at http://192.168.50.1:8081.

Accessing Kiwix

Open a browser to **http://192.168.50.1:8081**, or tap the Kiwix Library card on the dashboard. Kiwix has a built-in full-text search bar that searches across all installed content at once — type any symptom, procedure, or topic and results appear instantly, offline.

Content Catalogue

TIER	CONTENT	SIZE	BEST FOR
1	WikiMed Medical Encyclopedia	~471 MB	Mass-casualty events, medical support
1	Wikipedia (Mini)	~1.2 GB	General field reference, briefings
1	Wikivoyage	~820 MB	Maps, local facilities, travel info
2	iFixit Repair Guides	~2.3 GB	Equipment repair in the field
2	Wikipedia (Full EN)	~22 GB	Complete reference — large download

Adding a Custom ZIM

1. Download a ZIM from kiwix.org to a device with internet.
2. Copy it to the Pi: `scp mybook.zim fieldcommand@192.168.50.1:/opt/kiwix/zim/`
3. Register it: `sudo kiwix-manage /opt/kiwix/library.xml add /opt/kiwix/zim/mybook.zim`
4. Restart: `sudo systemctl restart kiwix`
5. Open http://192.168.50.1:8081 — the new content appears.

33 Offline Maps, GPS & Health Monitor

This chapter covers three infrastructure features that operate in the background to support the rest of FieldCommand: the offline map tile system that powers both the Tactical APRS Map and the Starcom Resource Map, the GPS live position feed that provides accurate coordinates across all tools, and the Health Monitor that tracks system vitals and service status in real time.

32.1 Offline Map Tile System

Map tiles are downloaded in advance, stored as MBTiles files on the Pi, and served locally on port 8083. Both the Tactical APRS Map and Starcom Resource Map use offline tiles by default. The default source is USGS Imagery+Topo Hybrid — public-domain satellite imagery with roads, contours, and place names.

Downloading Tiles

1. Run `sudo bash /opt/fieldcommand/scripts/download_tiles.sh` for the interactive menu (106 presets: all Illinois counties, WI/IN/IA border counties, all 50 states).
2. Search presets: `download_tiles.sh --search "mchenry"`
3. Download a preset: `download_tiles.sh --area "your county IL" --zoom 8-16`
4. Tiles are stored in /opt/fieldcommand/data/tiles/ and served by the tile server on port 8083.

32.2 GPS Live Position

If a USB GPS receiver is connected to the Pi (via the powered USB hub), gpsd provides live coordinates to the tactical map, health monitor, and NWS alerts. The GPS receiver creates a stable **/dev/gps0** device via udev rules installed automatically.

CHECK	COMMAND
GPS status	gpspipe -w -n 5
GPS device	ls -la /dev/gps0
gpsd status	sudo systemctl status gpsd
Live fix data	cgps -s

32.3 Health Monitor

The health monitor runs as a background service on port 5051 and feeds the dashboard sidebar with live system diagnostics. Raw data is at **http://192.168.50.1:5051/health**.

METRIC	NORMAL	ACTION IF EXCEEDED
CPU Temperature	< 70°C	Improve ventilation. Pi 5 throttles at 85°C.
Memory Usage	< 80%	Close unused browser tabs; restart non-essential services
Disk Usage	< 90%	Archive old logs; remove unused ZIM files
Service Status	All green	Red dot = service down. Restart with systemctl.

METRIC	NORMAL	ACTION IF EXCEEDED
Internet	Connected (if avail.)	Check Ethernet. Core features work offline.
GPS	Fix (if attached)	No GPS is fine — coordinates configured at install.

Restarting a Stopped Service

```
sudo systemctl restart fcc-lookup # or: health-monitor, ics-platform, fieldcommand-refs, kiwix, deadmans
```

34 Network Hardware — ASUS RT-BE58 Go & UniFi Switch Lite 16

FieldCommand v1.0 uses a three-router mesh network rather than a single access point. The primary ASUS RT-BE58 Go travel router manages WAN connectivity, DHCP, and the EMCOMM-NET Wi-Fi access point. Two additional RT-BE58 Go routers operate as AiMesh nodes extending EMCOMM-NET to secondary rooms, outdoor staging areas, and upper floors — all sharing the same SSID and password with seamless device roaming. The Pi itself never broadcasts Wi-Fi. It communicates over wired Ethernet only.

DEVICE	ROLE	CONNECTION
ASUS RT-BE58 Go (primary)	Wi-Fi 7 AP · DHCP server · WAN gateway · AiMesh controller	WAN: the cellular modem · USB WAN: the satellite dish failover · LAN: UniFi switch
ASUS RT-BE58 Go (mesh node 1)	EMCOMM-NET extension — same SSID, seamless roaming	Wired backhaul via UniFi Switch Port 11
ASUS RT-BE58 Go (mesh node 2)	EMCOMM-NET extension — third coverage zone	Wired backhaul via UniFi Switch Port 12
UniFi Switch Lite 16 PoE	16-port GbE managed switch · 8x PoE · wired distribution hub	Port 1: router uplink · Ports 2-12: devices · Ports 13-16: spare

33.1 EMCOMM-NET Wi-Fi Coverage

All three routers broadcast EMCOMM-NET on 2.4 GHz and 5 GHz simultaneously. Devices connect to whichever band and router provides the strongest signal and roam automatically as operators move around the venue. A single RT-BE58 Go covers approximately 2,000 to 2,500 square feet indoors. With both mesh nodes active, the system covers 7,500 to 20,000 square feet depending on building construction. For outdoor SAR staging areas, a node on battery at the perimeter extends EMCOMM-NET to field positions without additional cabling.

33.2 WAN Connectivity — the cellular antenna unit Primary + the satellite dish Failover

The ASUS router manages two WAN sources with automatic failover. the cellular modem (your cellular carrier and your cellular carrier dual-carrier) is always the primary. satellite internet is secondary — the ASUS switches to it automatically within 30 to 60 seconds when cellular drops, and switches back when cellular recovers. No operator action is required for the failover in either direction.

PRI	WAN SOURCE	CONNECTION	ACTIVATES
1	the cellular antenna unit (your cellular carrier + your cellular carrier)	PoE Ethernet from Drum or Switchblade to ASUS WAN port	Always — default for all activations
2	satellite internet	the satellite dish Ethernet adapter + USB-Ethernet to ASUS USB WAN	Auto when cellular drops
3	EOC site Ethernet	Site cable to ASUS WAN port	Manual — when site internet available
4	USB tether or venue Wi-Fi	Phone USB to ASUS USB port or ASUS WISP mode	Manual — emergency fallback

33.3 the cellular antenna unit Antennas

FieldCommand carries two the cellular antenna unit antennas to handle different deployment sites. Both connect via a single outdoor-rated PoE Ethernet cable to the ASUS WAN port. The modem is integrated into the outdoor antenna enclosure — no separate modem box is needed. Power travels up the same cable that carries data.

ANTENNA	USE WHEN	AIMING
Drum (omni 5G/LTE)	Default — every activation. Mounts on any mast, tripod, or vehicle roof rack.	No aiming needed — picks up all towers equally.
Switchblade (directional)	When Drum signal is poor. Folds flat for transport. Swap the PoE cable from Drum to Switchblade on the same ASUS WAN port.	Aim toward nearest tower using the cellular antenna unit app. Rotate slowly — stop at peak signal.

■ the cellular antenna unit data plan management: Keep the plan in Standby mode ($5/month) between activations. Standby maintains the SIM and multi-network capability so you can activate at full speed the moment an emergency is declared. Activate and pause plans at instyconnect.com. Active plan cost is approximately $79-99/month on the Multi-Network Unlimited plan.

33.4 WAN Status Dashboard

The WAN Status Dashboard at http://192.168.50.1/wan-status.html gives operators a live view of all WAN sources from any browser on EMCOMM-NET. The active WAN source is shown in a large color-coded panel: green for the cellular modem, blue for satellite internet, red for offline. the cellular antenna unit signal strength, carrier, and technology are displayed. the satellite dish latency, throughput, and obstruction percentage appear when the dish is active. A connectivity test panel verifies reachability to NWS weather alerts, APRS-IS, Cloudflare DNS, and the AMPRNet gateway every 60 seconds. A WAN event log records every failover and failback with UTC timestamps.

33.5 AMPRNet / 44Net Gateway

A dedicated Raspberry Pi 5 at 192.168.50.2 runs the AMPRNet gateway service. It maintains a permanent WireGuard encrypted tunnel to amprgw.ampr.org and routes the entire 44.0.0.0/8 AMPRNet address block for all EMCOMM-NET devices. This means any device on EMCOMM-NET — phones, tablets, operator workstations, and the FieldCommand Pi itself — can reach other AMPRNet stations globally without routing traffic through the commercial internet.

AMPRNET USE CASE	HOW IT WORKS
Winlink via AMPRNet	Pat Winlink connects to Winlink RMS gateways on 44.x.x.x directly over AMPRNet — bypassing commercial internet when only amateur radio paths are available.
APRS-IS via AMPRNet	Graywolf and YAAC connect to APRS-IS servers on 44.x.x.x — keeping APRS traffic within the amateur radio network.
Inter-node FieldCommand	If a second your organization FieldCommand system with its own 44Net gateway is deployed, the two systems can share data over AMPRNet without commercial internet routing.
Direct station connectivity	Any amateur station worldwide with a 44.x.x.x address is directly reachable from any EMCOMM-NET device for data exchange or status checking.

The gateway status page at http://192.168.50.2:9000 shows tunnel state, last handshake time, AMPRNet address, bytes transferred, and system health of the gateway Pi. Access requires a valid FCC amateur radio callsign. The FieldCommand dashboard health monitor also shows a 44Net status indicator updated every 30 seconds by the amprgate-poll background service.

33.6 AMPRNet Gateway — Access Control

Access to the AMPRNet gateway is restricted to FCC-licensed amateur radio operators. This is both a security measure and a Part 97 compliance requirement. The gateway uses a two-level access model:

HOW TO ACCESS	WHAT YOU CAN DO
Status Dashboard (any EMCOMM-NET device) — Open http://192.168.50.2:9000. Enter your FCC callsign at the login prompt. Validated against the local FCC database on the FieldCommand Pi.	View tunnel state, AMPRNet address, last handshake, traffic stats, routes, and gateway health. Read-only — no tunnel control.
Tunnel Control (gateway Pi keyboard only) — Open Chromium to http://localhost:9001 on the gateway Pi itself. Log in with FCC callsign. Port 9001 is blocked from the network — physical presence required.	Bring tunnel up / down / restart. All actions logged with callsign, timestamp, and IP address.

■ The /api/status endpoint on port 9000 does not require authentication. This is intentional — the FieldCommand Pi poll service reads it every 30 seconds as a machine-to-machine call to update the dashboard WAN indicator. This endpoint is read-only and returns no sensitive data. All web UI access and all tunnel control actions are logged to /var/log/amprgate-access.log on the gateway Pi.

33.7 Network IP Reference

DEVICE	IP ADDRESS	ADMIN URL
FieldCommand Pi 5	192.168.50.1	http://192.168.50.1
44Net Gateway Pi 5	192.168.50.2	http://192.168.50.2:9000
ASUS RT-BE58 Go (primary router)	192.168.50.254	http://192.168.50.254
Windows Laptop (recommended)	192.168.50.3	DHCP reservation in router
Color MFP Printer (recommended)	192.168.50.10	DHCP reservation in router
Pi 500 Workstations (×4)	192.168.50.20–23	DHCP reservations in router
the cellular antenna unit modem	10.1.1.1	http://10.1.1.1 (via the cellular antenna unit Wi-Fi)
the satellite dish dish admin	192.168.100.1	http://192.168.100.1 (via the satellite dish network)
Field devices (DHCP)	192.168.50.100–200	Assigned automatically by ASUS router
CUPS print server	192.168.50.1	http://192.168.50.1:631
Kiwix offline library	192.168.50.1	http://192.168.50.1:8081
Pat Winlink	192.168.50.1	http://192.168.50.1:8090
Health Monitor API	192.168.50.1	http://192.168.50.1:5050/health

DEVICE	IP ADDRESS	ADMIN URL
WAN Status Dashboard	192.168.50.1	http://192.168.50.1/wan-status.html
AMPRNet Gateway Status	192.168.50.2	http://192.168.50.2:9000

35 JS8Call — HF Digital Keyboard Messaging (Windows)

http://192.168.50.1/js8call

JS8Call is a keyboard-to-keyboard HF digital messaging mode based on FT8 technology. It is designed for very weak signal conditions — stations can exchange readable messages at signal levels 20 dB below what voice would require. JS8Call supports direct messaging between two stations, group messaging to a named group callsign such as @EMCOMM, store-and-forward relay through intermediate stations when no direct path exists, and heartbeat beacons that advertise a station presence automatically.

JS8Call runs on the Windows laptop alongside Winlink Express, both connected to the IC-7300 via the single USB cable. The FieldCommand dashboard provides a purple JS8Call quick-launch card that opens JS8Call built-in web interface from any device on EMCOMM-NET — operators at the EOC table can monitor JS8Call traffic from their phones or tablets without being at the Windows laptop.

34.1 Recommended JS8Call Frequencies

BAND	DIAL FREQUENCY	NOTES
40m (primary for ARES/RACES)	7.078.0 MHz	Primary JS8Call calling frequency — most activity
80m (regional, nighttime)	3.578.0 MHz	Good regional coverage after dark
20m (long distance)	14.078.0 MHz	DX and national nets — good during daylight
17m (secondary)	18.104.0 MHz	Less congested alternative to 20m
60m (emergency designated)	5.357.0 MHz USB	FCC Part 97.303(h) emergency channels

34.2 JS8Call for EmComm

USE CASE	HOW TO DO IT
Direct station message	Type @[CALLSIGN]: followed by your message. The addressed station receives it even if they are not actively watching — JS8Call stores incoming messages.
Group message (all EMCOMM stations)	Type @EMCOMM: followed by your message. All stations monitoring the @EMCOMM group receive it and it appears highlighted in their message window.
Heartbeat beacon (announce presence)	Enable Heartbeat in JS8Call settings. Your station transmits a short beacon every 15 minutes advertising your callsign, grid, and signal strength.
Store-and-forward relay	If direct contact is not possible, any intermediate JS8Call station on the path can relay your message to the destination automatically.
Query station SNR	Send CALLSIGN SNR? to ask a station to report your signal strength. Useful for propagation assessment before attempting Winlink.

36 ICS Planning P — Operational Planning Cycle

http://192.168.50.1/ics/planningp.html

The Planning P page is an interactive guide to the ICS operational planning cycle. It presents all 15 phases of the Planning P — from the initial incident through each operational period — with the standard agenda, required ICS forms, and attendee list for every phase. Access it from the ■ **Planning P** tab on any ICS platform page.

The 15 Phases

PHASE	NAME	KEY FORMS
1	Incident / Event	ICS-201 (begin)
2	Notification	ICS-201
3	Initial Response & Assessment	ICS-201
4	Initial UC Meeting	ICS-201, ICS-202
5	Incident Brief — ICS-201	ICS-201
6	IC/UC Develop/Update Objectives Meeting	ICS-202
7	Command & General Staff Meeting/Briefing	ICS-202, ICS-203
8	Preparing for the Tactics Meeting	ICS-215, ICS-215A
9	Tactics Meeting	ICS-215, ICS-215A, ICS-203, ICS-204
10	Preparing for the Planning Meeting	ICS-204, ICS-205, ICS-206, ICS-207, ICS-208
11	Planning Meeting	ICS-202 through ICS-206
12	IAP Prep & Approval	ICS-202 through ICS-208 (complete IAP)
13	Operations Briefing	Complete IAP, ICS-204 by Division
14	Execute Plan & Assess Progress	ICS-214, ICS-209, ICS-309
15	New Ops Period Begins	ICS-214, ICS-209

Using the Planning P Page

1. Open the ICS Platform (http://192.168.50.1/ics/) and click the ■ **Planning P** tab.
2. The reference image of the official Planning P diagram is shown on the left for reference.
3. The 15 phase buttons are listed in the center panel, organized into five color-coded groups.
4. Click any phase button to see its details in the right panel.

Phase Detail Panel

SECTION	WHAT IT SHOWS
Standard Agenda	Numbered list of the standard agenda items for that meeting or activity
Required Forms	ICS form numbers as clickable chips — click to open the form directly
Who Should Attend	List of ICS roles required or optional for that phase, with red/green roster dots showing who is currently checked in

The Phase Color Groups

COLOR	GROUP	PHASES	ICS PHASE OF PLANNING
Gray	Initial Response	1–5	Stem — initial incident response
Yellow	Establish Objectives	6–7	Stem — incident objectives set by IC/UC
Red	Develop the Plan	8–11	Loop — tactics through planning meeting
Green	Prepare & Disseminate	12–13	Loop — IAP approval and ops briefing
Teal	Execute, Evaluate & Revise	14–15	Loop — field operations and next period

Generating a Briefing Sheet

1. Select any phase by clicking its button.
2. Click **Generate briefing sheet** in the detail panel.
3. A printable one-page cover sheet opens with the phase name, agenda, required forms, and an attendance table with signature lines.
4. Print it for the IAP package or the planning meeting folder.

> The Planning P page is a guide, not a workflow enforcer. You do not need to click through the phases in order. Use it as a quick reference during planning meetings to ensure nothing is missed — especially on activations where the planning cycle is compressed.

34 Appendix — Administration & Quick Reference

A1 Installation & Updates

COMMAND	WHAT IT DOES
sudo bash install.sh	Interactive installer — sets callsign, coordinates, Wi-Fi, services
sudo bash update.sh	Updates FieldCommand code and restarts services
sudo bash kiwix_setup.sh	Install/manage Kiwix offline content
sudo bash download_tiles.sh	Download offline map tiles

A2 Wi-Fi & Network

ITEM	DETAIL
SSID	EMCOMM-NET
Pi IP	192.168.50.1 (static, set via nmcli on eth0)
DHCP range	192.168.50.100 – 192.168.50.200
Router admin	http://192.168.50.1 (ASUS RT-BE58 Go)
CUPS admin	http://192.168.50.1:631 (printer management)

A3 Database Backup

The main database is a single SQLite file. Back it up by copying **fieldcommand.db** to a USB drive. Export the roster CSV monthly and keep a copy on a USB drive labeled FIELDCOMMAND as a secondary backup. Plugging in a drive labeled FIELDCOMMAND automatically triggers a full rsync backup of the application data via the udev backup rule.

A4 Winlink — Email Over Radio

Most deployments use **Winlink Express** as the primary Winlink client on the Windows laptop with the IC-7300 + VARA HF. The Pi also runs **Pat** as a backup browser-based Winlink client on port 8090. Never run both clients with an active session on the same radio simultaneously.

A5 Service & Port Reference

SERVICE	PORT	DESCRIPTION
nginx	80	Web server — serves all HTML pages
fcc-lookup.service	5050	FCC callsign API, net log, forms, DMS, roster, resources
health-monitor.service	5051	System health API
ics-platform.service	5055	ICS incident platform API
fieldcommand-refs.service	5056	Reference library API

SERVICE	PORT	DESCRIPTION
Graywolf APRS	8080	APRS client with REST API and WebSocket
Kiwix	8081	Offline library — WikiMed, Wikipedia, iFixit
YAAC APRS	8082	Secondary APRS client
Tile server	8083	Offline map tile server (MBTiles)
Pat Winlink	8090	Browser-based backup Winlink client
CUPS printer	631	Print server — USB printer shared to EMCOMM-NET
JS8Call (Windows)	2442	JS8Call TCP API on Windows laptop